Ajay Devidas Wanekar

M.Tech Energy Science and Technology - TA

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SUMMARY

M.Tech student at IIT Hyderabad with 2+ years of research experience in Artificial Intelligence and Machine Learning, specializing in Computer Vision, OCR-based text and character recognition, semantic search, and Vision-Language Models. Experienced in data preparation, image preprocessing, dataset annotation, model benchmarking, and building multi-tenant AI applications that match extracted text against reference data.

Degree	University/Institute	Year	CGPA/ Marks(%)
M.Tech Energy Science and Technology - TA	IIT Hyderabad	2027	6.8
B.E. Electrical Engineering	Savitribai Phule Pune University	2022	7.28
XII (Maharashtra State Board)	Vasantrao Naik Jr. College of Science and Arts, Cidco, Ch. Sambhajinagar, Maharashtra	2017	67.38%
X (Maharashtra State Board)	Maharashtra Public School, DTP, Ch. Sambhajinagar	2014	83.00%

WORK EXPERIENCE

Project Associate | LFOVIA Lab, IIT Hyderabad Sep 2024 – Jul 2025

Evaluated and fine-tuned Vision-Language Models (VLMs), including Florence-2 Base/Large, Llama 3.2 Vision, and Qwen2-VL, for multimodal text-image understanding and OCR-based text and character recognition, and benchmarked task-specific models (RetinaFace, DeepFace, Siamese networks) for detection, face analysis, and visual tracking.

Junior Research Assistant | TIHAN, IIT Hyderabad: Nov 2023 – Jul 2024

Worked on multimodal maritime perception by developing and curating near-infrared and sonar datasets, implementing sensor-specific preprocessing, annotation, data-quality optimization, and model-training pipelines for robust object detection in complex maritime environments, resulting in an IEEE conference publication.

Research Intern | WINET Lab, IIT Hyderabad: Jul 2023 – Nov 2023

Developed foundational sonar-based underwater object-detection datasets by implementing annotation, Python-based preprocessing, data augmentation, and markup pipelines to improve data quality and model robustness, with the work published at the IEEE International Conference on Sustainable Technology and Engineering (i-COSTE 2023).

PROJECTS

Document Text Recognition and Visual Understanding with VLMs (LFOVIA Lab, IIT Hyderabad):

- Evaluated and fine-tuned Vision-Language Models (Florence-2, Qwen2-VL, Llama 3.2 Vision) for OCR-based text and character recognition and multimodal text-image understanding, as alternatives to conventional task-specific models.
- Applied the same models to object detection, segmentation, crowd analysis, and action recognition, and benchmarked them in Python pipelines (precision/recall, IoU).
- Curated and annotated RGB and depth-mapping datasets with preprocessing, labeling, and quality-validation pipelines.

AI Search and Multi-Tenant Conversational Platform (IIT Hyderabad Build Program):

- Built a multi-tenant generative-AI platform (voice, chat, email) with a hybrid semantic + keyword retrieval pipeline that matches noisy, inconsistently phrased text against a reference knowledge base, with LLM-based intent classification, improving answer relevance by 18%.
- Trained and tested various voice models (speech-to-text, text-to-speech) for the platform's multilingual voice channel.

THESIS PROJECT

Machine Learning for Non-Oxidative Methane Coupling in a DBD Reactor

- Developed and benchmarked Linear Regression, Random Forest, XGBoost, and Gaussian Process Regression models to predict reactor performance from operating conditions, validated with Leave-One-Out Cross-Validation (LOOCV), reaching cross-validated R^2 up to 0.996 across six reactor-performance outputs, and identified Bayesian Optimization for experiment-guided process optimization.
- Recently adapted the RRSI method to this work to rigorously verify that model improvements are real.

TECHNICAL SKILLS

Programming & Software: Python, SQL, Linux, C++, PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, NumPy, Pandas, Scikit-learn

OCR & Computer Vision: OCR (Text and Character Recognition), Image Preprocessing, Object Detection, Segmentation, Data Annotation & Augmentation, CNN, YOLO

Semantic Search & LLMs: Hybrid Semantic + Keyword Retrieval, RAG, LLM Intent Classification, Vision-Language Models (Florence-2, Qwen2-VL, Llama 3.2 Vision), LLM Fine-Tuning, Transformers

Data & Evaluation: Data Preparation, Dataset Curation, Precision, Recall, IoU, Cross-Validation (LOOCV), Model Benchmarking

CONFERENCE PUBLICATIONS

[1] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. "Novel Approach to Underwater Object Detection Using Sonar Sensors for Autonomous Underwater Vehicles (AUVs)." IEEE i-COSTE 2023, pp. 1–5.

[2] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. "Object Detection and Classification for Autonomous Surface Vehicles (ASVs) Through Near-Infrared Imaging." IEEE i-COSTE 2023, pp. 1–6.

HACKATHONS, ACHIEVEMENTS & LANGUAGES

- Top 10 All India - Winner, Meta x Pragati AI Hackathon.
- AI and X Hackathon, IIT Hyderabad participant

Languages: English, Hindi, Marathi, Japanese (Basic)